

AISSAOUI Ismail

AI & Data Science Engineer

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PROFILE

AI & Data Science Engineer specialized in Human-Computer Interaction (HCI) and intelligent system architecture. Focused on designing new interaction paradigms that treat GenAI as both a **material** and a **medium** for mediated action. Expert in building prototypes that explore the redistribution of agency, control, and responsibility between humans and agentic systems.

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, Lightning, **Agentic Architectures**

AI & HCI Techniques: Interaction Design, **Mediated Action**, GenAI as Medium, Prototyping

Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization

Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS

Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy

Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

Research & HCI: User Studies, Qualitative/Quantitative Methods, Statistics, Human-Centered Design, Mediated Action

Soft Skills: Technical leadership, Scientific communication, Interdisciplinary collaboration

PROFESSIONAL EXPERIENCE

AI Engineer Intern — Digital Twin & Predictive Maintenance

2026

Sonatrach — Algeria

Developed a Digital Twin for gas turbines enabling real-time monitoring and predictive maintenance. Implemented advanced anomaly detection and RUL estimation using deep learning models.

Network Engineering Intern

Sep 2024

Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA)

2021–2026

Engineering and Master's Degree in Computer Science

Specialization: Artificial Intelligence & Informatics

KEY ACHIEVEMENTS

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- Developed custom Transformer models (BERT, GPT, RAG) for real-world multilingual applications.
 - Built explainable AI platforms for HR tech and public institutions — reducing bias and improving decision quality.
 - Designed a graph-based recommendation system with +24% accuracy improvement (dataset: 60K profiles).
 - Delivered complete ML pipelines: data ingestion, validation, training, CI/CD, and monitoring.
 - Built real-time vision systems (YOLO, ByteTrack) reducing latency by 32%.

PROJECTS

Spatial Interaction & Geo-RAG

2025

Architected a map-oriented assistant where GenAI acts as a mediating substrate. Addressed the **gulf of execution** by simplifying intent expression and implementing dynamic context maintenance using latent embeddings.

Tech: ChromaDB, LangChain, React — Concepts: Mediated Action, Context Maintenance, Medium

Aura — Enterprise AI Trading System

2025

Full-stack automated crypto trading platform powered by hybrid LSTM forecasting and **Reinforcement Learning** (Deep Q-Learning) to optimize entry/exit policies under long-term reward maximization. Includes advanced technical indicators, real-time dashboards, and interactive backtesting.

Tech: TensorFlow, Keras, FastAPI, Next.js, WebSockets — Concepts: LSTM, Reinforcement Learning, Deep Q-Learning, Time-Series Analysis

Advanced NLP Pipeline with Multi-Optimization

2025

Multilingual moderation optimized with PSO, Genetic Algorithm, and Bayesian Optimization.

Tech: DistilBERT, PyTorch, FastAPI — Concepts: Optimization, Transformers, Classification

Intelligent University Scheduler

2025

Automatic generation of conflict-free timetables using hybrid heuristics and evolutionary algorithms.

Tech: FastAPI, NumPy, Pandas — Concepts: GA, PSO, ACO, Simulated Annealing

Mini-GPT (Decoder-only)

2025

Implementation of a compact GPT model with controlled training and sequential text generation.

Tech: PyTorch, CUDA, AdamW, W&B — Concepts: Transformers, Language Modeling

North African Sentiment BERT

2025

LoRA adaptation of BERT for sentiment analysis in Maghrebi dialect.

Tech: PyTorch, LoRA, MLflow — Concepts: Fine-tuning, Low-Rank Adaptation

Geo-RAG — Map Assistant

2025

RAG system integrating geospatial data for map-oriented queries.

Tech: ChromaDB, GeoPandas, Streamlit — Concepts: RAG, Spatial Reasoning

Recognizini — Agency Redistribution

2025

Hybrid facial recognition system exploring responsibility redistribution through an **iterative feedback loop**. Users act "through" the system to correct and guide inference in real-time.

Tech: OpenCV, PyTorch, React — Concepts: Agency, Human-Centric AI, Control Architectures

Real-time PPE Detection (YOLOv8)

2025

Real-time detection of safety equipment (helmet, vest) for compliance monitoring.

Tech: YOLOv8, OpenCV — Concepts: Object Detection, Streaming

Traffic Analytics Suite

2025

Traffic analytics including tracking, speed estimation, and license plate recognition.

Tech: YOLOv8, ByteTrack, EasyOCR — Concepts: Tracking, OCR, Analytics

Hadj Management System

2024

End-to-end logistics platform for group management and itineraries.

Tech: React, FastAPI, PostgreSQL — Concepts: Workflow, Synchronization

Hybrid POS & E-Commerce System

2025

Offline-first POS system integrated with an online store.

Tech: PyQt, SQLite, PostgreSQL — Concepts: Offline/Online Sync, Inventory Management

LANGUAGES

Arabic: Mother Tongue | English: Advanced | French: Conversational